

Delirious Departures

A nomadic untethered VR experience

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ABSTRACT

Delirious Departures is a VR experience in the form of an installation/performance. It questions the nature of travelling and our relationship to others. It explores the desire and impossibility of travelling, of leaving meaningfully in times of pandemic. The participant moves through impressive environments, based on iconic Belgian railway stations. An actor acts as guide, antagonist and director of the journey. The mocaped actor mixes with canned animations, intelligent avatars and figures in fixed poses, in various degrees of abstraction. Who is real, what is live? The experience of the participant is visualized in an experimental XR set-up which projects a mix of realtime video and computer graphics. The one-on-one performance builds on animation and crowd simulation technology developed by Inria and Cubic Motion (Epic Games).

KEYWORDS

Nomadic VR, VR, XR, LIDAR, avatars, crowd simulation

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DELIRIOUS DEPARTURES

Delirious Departures is a nomadic untethered VR experience by Brussels based arts collective CREW. It combines intelligent avatars, a live actor and awe-inspiring environments.

Thematically, *Delirious Departures* questions the nature of travelling, of departing, our relation to strangers. It explores the desire and impossibility of travelling, of leaving meaningfully in times of pandemic. In the public space of the railway station most people are passers-by, absorbed in their own thinking, music, reading, daydreaming, smart devices. For some it is a work destination: they sell train tickets, coffee, sweets, magazines, they sweep the floor and the toilets, repair the stationary escalator, or invisibly announce the arriving trains. Others pretend to be travellers but are there to warm themselves or to temporarily withdraw from the outside world. Fragments of conversation, foreign languages, service announcements, footsteps, coffee slurping travellers and many other (inter)actions and sounds blend together. Human-like figures, static

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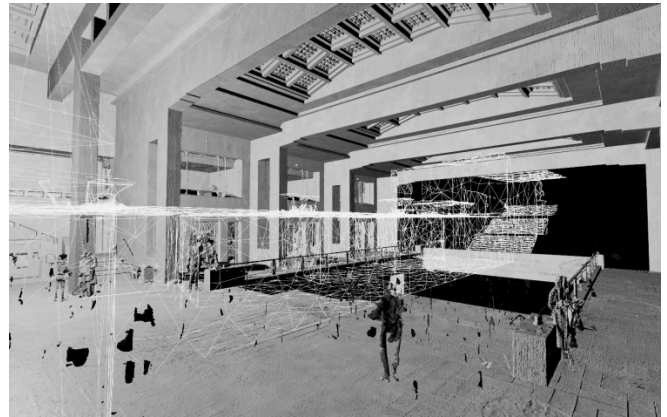


Figure 1: Impressive environment

or dynamic, prerecorded or live, question the nature of the self and the other.

Delirious Departures was first presented in the EUROPALIA TRAIN AND TRACKS festival. CREW latched into the theme of the festival by LIDAR scanning iconic Belgian railway stations. This happened during the covid pandemic. The various sanitary measures made the railway stations deserted. The few people who were present kept their distance and stayed mostly stationary. Normally people move around and are automatically removed from LIDAR scans by the reconstruction algorithm. During the lockdown, there were many people who kept their distance and stayed perfectly still. They are captured in the scans, which adds to the static, post-apocalyptic quality of the virtual world.

The scans are reconstructed, shaded and lit with a particular method, aiming for a specific aesthetic.

This method obtains impressive, awe-inspiring, static environments. These are offset by switching to simplified scenes, wherein there is a lot of movement. In these scenes, the person in immersion is confronted with agents with various degrees of interactivity. In this way we create a juxtaposition of static, highly detailed worlds, with minimal worlds only populated by moving agents, moulded by the omitted architecture.

The problem of the representation of humans in 3D graphics is as old as the field itself. CREW artistic research has led us to avoid any photorealism as uncanny valley cannot be crossed in VR with multiple agents. Similarly, CREW avoids cartoonish representations. Rather the artists approach the model of the human from a semi-abstract sculpture and fine arts perspective. Indeed, high quality animation lets us reduce the human form to a minimum while still being perfectly recognizable and relatable. Different stages of this abstraction are present in the experience.



Figure 2: Movement in immersion



Figure 3: XR example with tracked actor

An actor with Xsens full body tracking acts as both guide, antagonist and director. The actor controls the flow of the different scenes and can change his appearance in relation to the other avatars and the participant. The participant doubts which avatar is real or “live”. The actor communicates non-verbally with the person in the experience. Tracked objects as well as tactile interventions are used as mechanics to maximize immersion.

Essential to the experience is free movement. CREW has been working on embodied VR experiences since the early 2000s and has found movement to be essential to immersion, contrary to what is regularly shown at VR festivals. CREW collaborates with EDM (University of Hasselt) to develop custom tracking solutions. This version of the experience however uses standard steam VR tracking of maximum 10 by 10 meters, which is about the minimum for a meaningfully immersive experience. The interactions with the actor and the architecture of the worlds are devised in such a way that they stimulate movement.

The VR experience itself is limited to one person, 15 minutes. CREW aims to exteriorise the VR immersion. An experimental XR set-up is realized so that the experience can be projected and observed by the general audience in a mix of video and computer graphics.

The installation/performance is part of the Horizon 2020 research project Photoreal REaltime Sentient ENTity, with partners Framestore, Cubic Motion (Epic Games), Inria, University of Augsburg, Infocert, Brainstorm and University Pompeu Fabra. Within the context of the PRESENT project, CREW experiments both technically as conceptually with the role of humanoid representations in VR experiences. From the PRESENT project, *Delirious Departures* integrates two components. Interaction Fields is an algorithm for the UMANS crowd simulator by Inria. It is integrated in Unreal Engine. It is an artist-oriented tool that allows for quick scenarizing of complex group dynamics. The motion generator, developed by Cubic Motion (acquired by Epic Games) is an animation tool based on an implementation of motion matching, focused on natural locomotion instead of more action-oriented game motion.